

Yujian Zhang

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Education

Tianjin University - Master of Computer Science and Technology	Sep 2022 - Jun 2025
Rank 1/22 (87.41/100) Awarded Tianjin University Special Academic Scholarship , Bayesian Computing Scholarship, Gold Medal in National University Algorithm Design and Programming Challenge, etc.	
Sichuan University - Bachelor of Water Science and Engineering (Outstanding Graduate)	Sep 2018 - Jun 2022
Rank 1/58 (86.17/100) Awarded National Scholarship , National Inspiration Scholarship, Outstanding Graduate of Sichuan Province , Outstanding Communist Youth League Cadre, Excellent Student of Sichuan University, etc.	

Research

Research on Semi-Supervised Domain Generalization - Student 1st author	Sep 2022 - Sep 2023
Multi-level Augmentation Consistency Learning and Sample Selection for Semi-Supervised Domain Generalization Accepted by <i>2024 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2024)</i>	
• Setting: The model is trained using only a small amount of labelled data, and the test data is from a different distribution than the training data. • Methods: This paper expands the distribution of the training samples using image and feature level style augmentation, and explicitly constrains the domain invariance of features to improve the model generalization ability from theory. In addition, the paper proposes a new sample selection strategy to select high-quality samples to label for more efficient model training. The proposed method achieves SOTA results on the domain generalization datasets PACS and OfficeHome, and extends the application scenarios of the existing research.	

Cross Research Papers of Machine Learning and Water Purification - 1st author/Co-author	Sep 2019 - Jun 2022
Data analysis and knowledge mining of water treatment experimental results using classical machine learning methods. For example, we use linear regression to predict the concentration of novel pollutants in water. The related research has published 8 SCI papers, including 4 papers of Chinese Academy of Sciences Q1, 3 papers of first author, and currently I have more than 260 citations on Google Scholar.	

Internship

Baidu Online Network Technology (Beijing) Co., Ltd. -- AI Cloud Group - AI Algorithm Intern	Jun 2024 - Present
• Video multimodal large model anomaly detection task fine-tuning. Investigate the application of video-language multimodal large model in video anomaly detection, and use the existing data to verify the effectiveness of related methods and fine-tune them to explore the optimisation scheme. The main work includes scheme research, scheme selection, data production, model fine-tuning and so on. • Generalizability exploration on traffic detection models. Explore how to improve the generalization effectiveness of Transformer-based target detection models on different city data. I am responsible for the programme development, evaluation set construction, model training, and data enhancement scheme exploration. The proposed method improved mAP50 by 0.7%-11.0% in real scenario data. • Vehicle Multi-Classification Model 0-1 Development. Collected business data and network data to train the vehicle recognition model with an average accuracy of 95.7%. The main work includes confirmation of requirements, evaluation set construction, data cleaning, and cold start. In addition to this model, a variety of classification/detection models were involved, and a total of 6 business requirements were completed.	

- **Multimodal Large Models for Document Localisation in Complex Scenes.** A new dataset is proposed to segment documents in complex scenes. The research involves the lightweighting of multimodal large models and semantic segmentation, etc. I am responsible for investigating the effect of existing multimodal semantic segmentation large models and exploring the lightweighting scheme of the models.
- **Refinement of large model semantic segmentation.** We investigated and reproduced the existing SOTA semantic/instance segmentation model, and used post-processing methods to refine the large model segmentation results by combining the multi-scale contextual semantic information of images.

Publications

Yu M, **Zhang Y**, Li X, et al. Multi-Level Augmentation Consistency Learning and Sample Selection for Semi-Supervised Domain Generalization[C]//ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2024: 6550-6554. (**Student 1st author**)

Zhang Y, Zhou G, Yue J, et al. Enhanced removal of polyethylene terephthalate microplastics through polyaluminum chloride coagulation with three typical coagulant aids[J]. Science of the Total Environment, 2021, 800: 149589. (**1st author**)

Zhang Y, He L, Wang G, et al. Defluorination and regeneration study of lanthanum-doped sewage sludge-based activated carbon[J]. Journal of Environmental Chemical Engineering, 2021, 9(4): 105740. (**1st author**)

Li Z, **Zhang Y**, Wei X, et al. A comprehensive study on the performance and mechanism of microplastics removal by enhanced coagulation methods[J]. Journal of Water Process Engineering, 2023, 56: 104238. (**Equal 1st author**)

Xing X, **Zhang Y**, Zhou G, et al. Mechanisms of polystyrene nanoplastics adsorption onto activated carbon modified by ZnCl2[J]. Science of The Total Environment, 2023, 876: 162763. (2nd author)

Huang L, He W, **Zhang Y**, et al. Chitosan enhances poly aluminum chloride flocculation system removal of microplastics: effective, stable, and pollution free[J]. Journal of Water Process Engineering, 2023, 54: 103929. (3rd author)

Zhou G, Huang X, Xu H, et al. Removal of polystyrene nanoplastics from water by CuNi carbon material: the role of adsorption[J]. Science of The Total Environment, 2022, 820: 153190.

He L, Wang G, Zhang X, et al. Lanthanum-doped activated carbon derived from municipal sludge for enhanced defluoridation: characteristics and mechanism[J]. Water Science and Technology, 2020, 82(8): 1643-1652.

Yang Z, Luo Y, Yue J, et al. Activation of O2 by zero-valent zinc assisted with Cu (II) for organics removal: Performance and mechanism[J]. Journal of Hazardous Materials, 2022, 424: 127506.

Skills

- Deep learning/machine learning frameworks: Numpy, PyTorch, scikit-learn, and PaddlePaddle.
- Programming Language: Python, C++, Linux Shell.
- Others: Office, Origin, Adobe Premiere Pro, Cinema 4D.

Research Interest

My research interests include, but are not limited to: **deep model generalization, semi-supervised learning, computer vision, multimodal large models.**